

Chapter

Digital Integrity in the Gulf: Policy Innovations and Technological Tools to Combat Corruption in GCC States

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Abstract

Digital technologies are increasingly used to enhance transparency and accountability in public governance. In the Gulf Cooperation Council (GCC), governments have deployed e-procurement systems, open data portals, and artificial intelligence (AI) tools to modernize public administration and deter corruption. However, the impact of these technologies varies widely across the region. This chapter analyzes how legal frameworks, institutional structures, and civic conditions shape the effectiveness of digital anti-corruption tools in six GCC countries: the United Arab Emirates, Saudi Arabia, Qatar, Kuwait, Oman, and Bahrain. Using a comparative legal-policy approach, the study identifies the enabling conditions that determine whether digital innovation results in structural integrity reform or merely administrative efficiency. Findings suggest that while digital tools offer significant potential, their success depends on coherent legislation, inter-agency coordination, and inclusive governance.

Keywords: Gulf Cooperation Council, digital governance, anti-corruption, e-government, legal reform, public procurement, transparency, artificial intelligence, blockchain, civic engagement

1. Introduction

The Gulf Cooperation Council (GCC) states are often hailed as pioneers of digital transformation in the Arab region. Over the past decade, governments in Saudi Arabia, the United Arab Emirates, Qatar, Kuwait, Bahrain, and Oman have rolled out national digital strategies, e-government platforms, and data-driven public services that reflect their ambition to modernize governance and foster economic diversification [1–3]. Digital transformation is frequently framed in official discourse as a cornerstone of “post-oil” development models, aiming to reduce bureaucratic inefficiencies, streamline public services, and enhance competitiveness in the global economy.

However, this drive toward digitization coexists with persistent concerns over public integrity, elite capture, and institutional opacity in several of these same countries [4–6]. In recent years, global governance indexes and corruption risk assessments have continued to highlight the discrepancy between formal transparency initiatives and informal practices that limit accountability. As the region's governments invest heavily in technologies such as artificial intelligence (AI), blockchain, and open data infrastructures, critical observers have raised concerns about whether these tools truly disrupt entrenched patronage systems or merely digitize existing power asymmetries [5, 7].

This apparent paradox – technological innovation in governance coexisting with unresolved structural corruption risks – raises an important question: Can digital tools meaningfully strengthen integrity, accountability, and transparency in the public sector in the Gulf?

Answering this question requires moving beyond the technocratic narratives that dominate regional digital transformation agendas. While it is tempting to view digital tools as neutral or inherently modernizing, their governance outcomes are deeply shaped by the legal frameworks, institutional contexts, and sociopolitical dynamics into which they are embedded. In other words, technology is not deployed in a vacuum; it reflects policy choices, institutional incentives, and strategic interests that vary widely across and within GCC states [6, 8–10].

While previous research has examined the role of foreign aid in combating corruption in the Arab region, including in post-Arab Spring contexts [11], this chapter focuses on the emerging role of digital technologies as instruments of public integrity within the specific institutional and political fabric of the GCC. It explores how digital technologies are being used – or could be used more effectively – to combat corruption in the GCC. Drawing on cross-national examples and comparative legal-policy analysis, the chapter examines the enabling legal frameworks, institutional dynamics, and operational functionalities of selected digital integrity tools.

These include blockchain-based procurement platforms, AI for anomaly detection, and open data initiatives designed to enable public scrutiny of governmental activities. Although such tools have been promoted in national digital strategies, their success is far from guaranteed. Their impact varies significantly across the region and depends on multiple intersecting factors, including political will, legal clarity, civic space, inter-agency coordination, and institutional autonomy [7, 9, 12].

In many cases, the implementation gap – between what technology promises and what it delivers – remains wide. For instance, a digital platform may centralize procurement procedures and reduce manual inefficiencies, but if it does not disclose contract data or allow public oversight, its anti-corruption value is limited. Similarly, blockchain registries may provide tamper-resistant databases, but without independent auditing mechanisms, their transparency potential is curtailed. And AI tools may detect anomalies in financial flows, but if these insights are not acted upon by empowered oversight bodies, the tools become mere technocratic ornaments [5, 9, 13].

Rather than treat digitalization as a silver bullet, the argument advanced here is that gains in integrity depend less on technological adoption alone than on the coherence of legal frameworks, the autonomy of oversight institutions, and the degree of public and media engagement [5, 10, 12]. The real value of technology lies

in how it is integrated into systems of governance that allow contestation, deliberation, enforcement, and participation. Absent such integration, digital innovation may enhance administrative efficiency while leaving corruption risks intact – or even hidden beneath complex technical layers [14].

This chapter thus adopts a critical institutionalist lens, one that situates digital governance reforms within broader political economy structures. Rather than assessing digital tools solely based on their technical features, the analysis probes the legal and institutional enablers – or inhibitors – of effective anti-corruption outcomes. By examining concrete tools implemented in the six GCC countries and assessing their legal-institutional contexts, the chapter seeks to distinguish symbolic innovations from substantive reforms.

The chapter is organized as follows:

- Section 2 provides a contextual overview of digital governance trends and anti-corruption challenges in the GCC.
- Section 3 explores the most significant digital tools adopted in the region to promote transparency and accountability, including their strengths and limitations.
- Section 4 discusses structural and institutional constraints that limit the impact of these tools, such as legal gaps, institutional centralization, and restricted civic engagement.
- Finally, Section 5 outlines strategic pathways to embed integrity into digital governance frameworks across the Gulf, combining legal reform, institutional redesign, and civic participation.

Ultimately, the chapter contributes to a growing literature that cautions against technological solutionism in anti-corruption policy and instead advocates for a holistic approach – one that aligns digital innovation with the foundational norms of public accountability, legal transparency, and participatory governance [9, 15, 16].

2. Digital governance and the quest for integrity in the Gulf

The GCC countries have aggressively pursued digital transformation agendas, often framed as strategic levers for economic diversification, public sector efficiency, and citizen-centric service delivery [1–3]. National initiatives such as Saudi Arabia’s Vision 2030, Qatar’s TASMU program, and the UAE’s Digital Government Strategy are emblematic of this shift toward a digitally enabled state [1, 2]. These frameworks emphasize smart government, seamless services, real-time data integration, and digital-by-default delivery models as markers of modernization and competitiveness.

These strategies have translated into tangible infrastructure. The UAE ranks among the global leaders in e-government readiness and AI adoption in the public sector [15, 17], while Saudi Arabia has built one of the most expansive digital government ecosystems in the Middle East [13]. Qatar has developed sectoral data governance frameworks under TASMU, and Oman has invested in blockchain

registries and digital identification systems [3, 18]. At a structural level, digital governance is no longer a peripheral reform but a central axis of state modernization in the Gulf.

However, digital transformation does not inherently produce transparency or accountability. While e-government can reduce face-to-face interactions that foster petty corruption, it does not automatically constrain elite capture, policy opacity, or abuse of discretionary power [4, 5, 19]. In fact, global literature on ICT-enabled citizen engagement stresses that information technologies must be embedded in institutional ecosystems that enable feedback loops, accountability mechanisms, and legal enforcement [9, 16, 20]. In contexts where political authority is highly centralized and civic engagement remains limited, technology may become an instrument of technocratic management rather than democratic responsiveness [6, 12, 21].

This divergence between digital ambition and institutional effectiveness reveals a deeper implementation gap. While technical capabilities exist, their impact is mediated by power structures, legal ambiguity, and the limited autonomy of oversight actors [10, 22, 23]. In such contexts, the transformative potential of technology is often neutralized or reabsorbed into dominant administrative logics.

In the Gulf, efforts to digitalize governance have yielded impressive service delivery gains, such as online licensing, mobile identity authentication, and digital payment integration. For example, platforms such as *Etimad* (Saudi Arabia), *Sahel* (Kuwait), and *UAE Pass* offer streamlined access to services and e-procurement. Yet, these platforms often lack embedded transparency features, such as open contract data, public audits, or real-time red-flag alerts [13, 14, 24]. Many GCC platforms are designed for internal efficiency rather than external accountability. While they reduce transaction costs and enhance service speed, they seldom empower civil society or independent media to monitor public spending or expose malpractice.

In some instances, centralized data architectures reinforce executive control over information without creating mechanisms for public contestation or independent review. This is particularly visible in the limited availability of disaggregated, real-time data on public procurement, budgeting, and administrative decisions. In Saudi Arabia, for instance, while tender announcements are posted online, comprehensive data on contract implementation, amendments, or performance benchmarks are rarely available to non-state actors [24].

Moreover, the legal environment for open government remains underdeveloped in several GCC countries. Freedom of Information (FOI) laws are either absent or narrowly interpreted, and independent oversight institutions – such as anti-corruption commissions or supreme audit bodies – often lack operational autonomy and access to critical government data [22, 25, 26]. In most Gulf countries, there is no legal obligation for government agencies to disclose data beyond voluntary transparency portals, which themselves often lack quality assurance or periodic updates.

For example, while Bahrain and Kuwait have adopted some transparency measures, including limited FOI provisions and public procurement portals, the enforcement gap remains significant. Attempts to introduce broader FOI laws often collide with entrenched executive prerogatives and national security narratives, which are frequently invoked to limit public access to government-held information [27, 28]. These legal voids are not accidental; they reflect a broader reluctance to institutionalize mechanisms of horizontal accountability, which would allow citizens, journalists, and opposition actors to scrutinize public actions.

Furthermore, digital governance projects are often deployed without citizen co-design or participatory oversight mechanisms. Civic engagement is treated as optional, not foundational. As a result, even sophisticated digital platforms risk becoming technocratic shells that increase efficiency without enhancing trust or legitimacy. The risk is not simply inefficacy – it is the legitimization of digital authoritarianism, where technological tools are used to optimize centralized control while projecting a facade of modernization [21, 29, 30].

Another overlooked dimension is data privacy and cybersecurity governance, which remains fragmented or underdeveloped in most GCC states. The integration of biometric identity systems, cross-ministerial data sharing, and AI-based surveillance tools raises important concerns about data protection and due process rights [29, 31]. In the absence of strong privacy laws or independent data protection authorities, the expansion of digital governance can facilitate opaque data practices, unaccountable surveillance, and algorithmic discrimination [10, 17].

These tensions highlight that digital governance, when deployed without legal safeguards or participatory mechanisms, can entrench rather than disrupt centralized control. The promise of digital transformation becomes conditional on governance models that privilege transparency, pluralism, and civic engagement – not just service optimization.

In sum, while the GCC has made considerable strides in building digital infrastructure, its impact on public integrity remains uneven and context-dependent. What is at stake is not the capacity to digitize, but the political willingness to democratize digital governance. The next section explores how these contradictions manifest in concrete tools – such as e-procurement platforms, blockchain registries, AI-based risk detection, and open data portals – and what they reveal about the region's evolving integrity landscape.

3. Digital tools for public integrity: Innovations in the Gulf

Governments in the Gulf have deployed a range of digital tools to enhance transparency, prevent corruption, and promote public integrity. These initiatives vary in scope, technological sophistication, and legal grounding. They reflect not only technological ambition but also political choices about what kinds of accountability are prioritized – and which are systematically avoided.

Despite the visibility of digital platforms and innovation labs, the use of technology for anti-corruption purposes remains uneven and often fragmented across the region. Tools such as e-procurement platforms, blockchain-based registries, AI applications, and open data portals offer significant potential; yet, their actual contribution to integrity depends on how they are designed, governed, and embedded in institutional ecosystems [5, 9, 12].

The following subsections examine the most prominent digital tools adopted in the GCC, highlighting both their innovative features and operational limitations.

3.1 e-Procurement and smart contracting

Public procurement remains one of the most vulnerable areas to corruption globally, and the GCC is no exception [4, 5, 19]. Procurement-related corruption can take many forms: bid rigging, collusion, kickbacks, ghost vendors, or conflicts of

interest. In response, several Gulf states have adopted digital procurement platforms to automate and monitor the tendering process.

Saudi Arabia's *Etimad* platform, launched under the Ministry of Finance, centralizes bidding processes, vendor registration, and payment management for all government entities [13]. It enhances procedural transparency and reduces opportunities for arbitrary allocation. However, detailed contract data and real-time red-flag analytics are not publicly accessible, limiting external scrutiny [13, 24]. *Etimad* also lacks user-friendly dashboards for non-expert stakeholders, such as journalists or watchdog NGOs.

Although *Etimad* has improved internal efficiency and reduced paper-based inefficiencies, it does not offer open access to audit trails or contract performance data – key elements for external accountability. The lack of interoperability with independent oversight institutions further limits its potential as a corruption-deterrence mechanism [13].

In Kuwait, the Central Agency for Public Tenders (CAPT) has introduced digital bidding modules, including electronic document submission and prequalification systems. However, procurement-related decisions – especially in strategic sectors such as defense, infrastructure, and hydrocarbons – are often exempt from full disclosure, reducing the integrity gains of digitalization [22, 23, 31].

Qatar's Ministry of Finance operates a Procurement Regulatory Authority portal that manages bids and contracts. While technically functional, access is generally restricted to registered firms. There is no public registry of procurement outcomes, no feedback loops, and limited user data on the identity of bidders, contract amendments, or delivery performance [3, 28].

By contrast, Oman's eTendering portal has introduced some transparency features, including downloadable RFPs and contract award announcements. Yet, here too, contract execution data, complaint mechanisms, and dispute outcomes remain opaque. In Bahrain, the Government Tender Board portal is among the most open, offering tender notices, bidder information, and award decisions. However, it lacks longitudinal data on implementation, subcontracting, or sanctions for non-compliance [3, 14].

These examples illustrate a broader pattern: digital tools are deployed to streamline bureaucratic processes but rarely to empower independent monitoring or civic scrutiny. The result is a form of partial digitalization that avoids politically sensitive reforms. Notably, none of the GCC countries have adopted Open Contracting Data Standard protocols, which provide detailed guidelines for publicizing each stage of the procurement process [32].

In this regard, the experience of Ukraine's *ProZorro* platform offers a useful counterpoint: it combines real-time publication of procurement documents with civil society oversight and algorithmic alerts for anomalies – elements still largely absent in the Gulf [33].

3.2 Blockchain for secure registries

Blockchain technology has been hailed as a revolutionary tool for transparency, offering tamper-resistant ledgers that can increase traceability and deter fraud. In the GCC, blockchain-based tools are emerging in land and asset registries, though adoption remains pilot-based or limited to specific government departments.

The Dubai Land Department has adopted blockchain to manage ownership transfers, aiming to reduce fraud and streamline real estate transactions [18]. The system ensures that each property record is time-stamped, encrypted, and stored in a decentralized format. Similarly, Bahrain has piloted blockchain applications for digitized government documents in partnership with private sector actors [3, 14].

However, while these systems promise tamper-proof recordkeeping, their impact on corruption depends on public accessibility and auditability. In most cases, blockchain platforms in the Gulf remain government-facing, without mechanisms for civil society or independent bodies to verify transactions or detect anomalies [18, 21].

For example, the Dubai blockchain registry allows internal verification but does not publish anonymized records or permit public comparison of ownership histories. This limits its usefulness as an anti-corruption tool and reflects an institutional design focused on efficiency rather than external transparency [18, 25].

Furthermore, the legal frameworks surrounding blockchain use remain vague, particularly regarding data ownership, consent, and algorithmic accountability. Without robust FOI laws or independent oversight institutions, the benefits of blockchain for transparency may remain largely theoretical or symbolic [5, 31].

In addition, blockchain systems are often developed through public-private partnerships, raising concerns about vendor lock-in, cybersecurity vulnerabilities, and the transparency of procurement arrangements for the underlying infrastructure [10, 29].

Recent research by Alshhadat (2024) confirms that regulatory ambiguity, cultural resistance, and insufficient blockchain literacy within public institutions remain major barriers to effective adoption in the GCC. These constraints contribute to a form of digital stagnation, in which blockchain is cited as a priority in national strategies but rarely implemented as a governance tool [34].

3.3 AI and predictive analytics

Artificial intelligence is being increasingly integrated into administrative systems across the GCC. Governments use AI tools for a variety of purposes: detecting anomalies in tax declarations, predicting procurement fraud, analyzing public service delivery efficiency, and even profiling customs risks.

In the UAE, AI is used to detect discrepancies in tax declarations, monitor performance indicators in health and education, and support digital identity authentication [15, 17]. Saudi Arabia has invested in predictive analytics and machine learning models to audit public procurement and internal expenditures, though little is known about their architecture or governance [13, 35].

While these tools can uncover hidden patterns and flag potential misconduct that manual processes may miss, they raise profound concerns about algorithmic opacity. Few of these systems are subject to algorithmic audits, and there is no publicly available information on how detection models are trained or evaluated [10, 29]. In opaque governance systems, AI tools risk reinforcing arbitrary decision-making rather than challenging it.

Moreover, most Gulf countries lack legal frameworks governing the ethical use of AI in the public sector. Issues such as explainability, accountability, and due process rights for individuals flagged by AI systems are rarely addressed [17, 29]. Without safeguards, there is a risk that AI will simply automate opacity and institutionalize bias [9, 29].

To function as effective anti-corruption tools, AI systems must be governed by transparent algorithms, subject to ethical oversight, and integrated into systems where human redress and appeal mechanisms exist. In the absence of civic oversight and legal clarity, the deployment of AI may expand control without enhancing integrity [23].

A recent comparative study by Albous et al. (2025) found that AI deployment in the Gulf often lacks algorithmic transparency and citizen oversight, and remains concentrated in executive agencies without cross-sectoral consultation or independent audits [36].

Similarly, Ashal and Murshed (2024) observed that, while AI-driven procurement surveillance has been piloted in Qatar and Oman, these tools are rarely subjected to external verification and remain dependent on internal governance rules that vary across agencies [37].

3.4 Open data platforms and civic participation

Several GCC states have launched open data portals to share government information with the public. These include the UAE's *Federal Open Data Portal*, *Qatar Open Data*, and *Saudi Arabia's National Data Management Office* [2, 3, 20]. The idea is to enable third-party actors – researchers, journalists, civil society organizations – to access data that can be used to monitor governance outcomes.

These initiatives mark a positive step toward transparency, yet their impact remains limited by the quality, granularity, timeliness, and usability of the data provided [22, 38]. In many cases, datasets are outdated, not disaggregated by ministry or region, and published in formats that are not machine-readable. Metadata, methodology, and update cycles are rarely disclosed, reducing their analytical utility.

Moreover, the absence of strong right-to-information laws and independent media restricts the ability of civic actors to use this data for accountability purposes [25, 38]. Most portals lack application programming interfaces (APIs) or feedback tools, and few support civic participation mechanisms, such as data requests or corrections.

Although these platforms present themselves as instruments of open government, they often serve symbolic rather than substantive functions. Public dashboards exist but rarely support civic feedback loops, co-creation mechanisms, or journalistic investigations [16, 30].

For example, Saudi Arabia's Open Data Platform publishes high-level statistics but does not enable users to track public budgets, follow procurement pipelines, or access sanction data. Kuwait's transparency initiatives are hindered by the lack of FOI legislation and the weak protection of whistleblowers [26].

The broader risk is that open data becomes decoupled from political contestation – a repository of information with no legal or institutional mechanisms to trigger change. Transparency without action, or without an audience, becomes performative. In the absence of legal mandates and civic capacity, open data platforms risk legitimizing state control rather than enabling democratic scrutiny [9, 16, 20].

A comparative summary of these tools across GCC countries is provided in Annex A. To complement this overview, we also present, in Annex B, a new synthesis table

comparing the legal frameworks, institutional mandates, and civic access mechanisms associated with each type of tool.

Moving forward, more granular case studies – detailing how specific tools were implemented, their political limitations, and pathways for reform – are essential to bridge the implementation gap and inform context-sensitive recommendations.

4. Structural constraints and governance gaps

Despite their digital ambitions, GCC countries face persistent structural and institutional barriers that limit the effectiveness of anti-corruption technologies. These constraints reflect not only technical challenges but also deeper issues of legal design, political centralization, and restricted civic oversight.

As previous findings suggest, even well-intentioned foreign aid can inadvertently reinforce corruption when institutions are weak and civic engagement is constrained [11]. Similarly, digital anti-corruption tools – no matter how advanced – are unlikely to succeed in the absence of legal accountability, institutional autonomy, and public participation [5, 9].

4.1 Legal gaps and asymmetric transparency

Legal frameworks for transparency and accountability remain uneven across the GCC. FOI laws are either absent (e.g., Saudi Arabia, UAE) or nascent, with limited scope and enforcement (e.g., Kuwait, Bahrain) [22, 25, 26]. Where FOI legislation exists, it is often undermined by broad national security exemptions, ambiguous procedures, and weak institutional capacity to process requests [27].

This legal vacuum is not simply a result of underdevelopment but a deliberate institutional choice. As several observers note, information control remains a pillar of regime stability in many Gulf monarchies, where public knowledge of elite networks, procurement decisions, or public assets is considered politically sensitive [16, 21, 23].

Moreover, national anti-corruption strategies frequently lack binding legal mandates or enforceable timelines. For instance, Qatar's 2021 National Anti-Corruption Strategy articulates broad goals – such as fostering transparency and ethical conduct – but does not establish enforceable obligations for ministries to publish procurement or audit data [28].

Across the region, open data platforms function in legal vacuums, with no obligation for accuracy, timeliness, or completeness. Governments retain full discretion over what is disclosed and when, undermining the possibility of independent verification or civic enforcement [9, 38].

This leads to a paradox: digital transparency without legal entitlement, where public access to information is voluntary, partial, and revocable. Without a legal right to know, citizens and oversight institutions are relegated to passive observers of state-selected narratives.

Recent research by Albous et al. (2025) shows that AI deployment across GCC countries occurs largely in the absence of enabling legislation or mandatory algorithmic disclosure rules. Regulatory frameworks on public-sector AI use remain vague or aspirational, with little clarity on citizen redress mechanisms or auditing

procedures [36]. This weak legal environment risks institutionalizing algorithmic opacity and further insulating digital tools from accountability.

Similarly, Alshhadat (2024) notes that, in the case of blockchain-based registries and procurement tracking systems, legal uncertainty – especially around data sovereignty and ledger governance – has prevented ministries from mainstreaming these tools beyond limited pilot projects [34].

4.2 Institutional centralization and executive control

GCC digital initiatives are often developed and managed by central ministries, royal courts, or executive councils, reflecting a model of top-down digitalization. While this enables rapid implementation, it also limits inter-agency coordination, weakens institutional checks, and shields platforms from independent scrutiny [12, 23].

Supreme Audit Institutions (SAIs) – where they exist – often report directly to the executive, limiting their operational autonomy. Many SAIs in the Gulf lack mandates to audit digital platforms, procurement algorithms, or blockchain infrastructure. In some cases, their access to data is restricted by “national interest” clauses that protect ministries from external audits [27, 35].

Transparency units or anti-corruption bodies, such as Nazaha in Kuwait, frequently lack prosecutorial power, budgetary independence, or secure tenure for leadership. This restricts their ability to monitor, investigate, or sanction malfeasance – particularly in high-profile or politically sensitive sectors [23, 31].

Furthermore, most GCC states lack interoperable platforms that allow oversight actors to access real-time public finance or procurement data. Even when tools exist, such as analytics dashboards, their use is generally restricted to the executive branch, excluding parliaments, courts, or civil society.

The resulting institutional architecture is hierarchical and insulated, where digital platforms are technically functional but legally disconnected from mechanisms of contestation, accountability, or reform.

According to Ashal and Murshed (2024), AI fraud detection systems in Gulf public procurement are rarely connected to independent oversight bodies or cross-institutional review. Instead, anomaly detection is handled internally, without transparent procedures for escalation, investigation, or sanction [37]. This reinforces verticalism in governance and weakens the potential of AI tools to catalyze structural accountability.

4.3 Limited civic engagement and media constraints

Public participation in digital governance remains minimal to non-existent across the region. Civil society organizations are tightly regulated, and independent investigative journalism is often constrained by defamation laws, licensing requirements, and state censorship [30, 38].

Without robust civic pressure, even well-designed digital systems can remain opaque, underutilized, or abandoned. For instance, Bahrain’s early experiments with civic hackathons under the “OpenGov” initiative were discontinued, and public access to dashboard APIs was subsequently restricted [3, 14].

In Saudi Arabia, while the *Etimad* platform allows businesses to monitor their own tenders, there is no interface for journalists, NGOs, or citizens to track

irregularities or raise red flags [13, 24]. Similarly, Qatar's open data portal lacks a grievance mechanism or participatory design component, reinforcing its top-down character [3, 20].

In environments where civil society is treated with suspicion and civic space is restricted, digital tools are orphaned from oversight. What remains is a model of "technocratic transparency" – governance without participation [9, 16].

Moreover, the absence of digital literacy programs, capacity-building initiatives, and legal protections for whistleblowers discourages citizen involvement. The chilling effect of surveillance and digital repression – including social media monitoring and cybercrime laws – further deters civic engagement with digital integrity tools [21, 30].

As noted by Alshhadat (2024), citizen-facing blockchain initiatives in the Gulf have suffered from poor uptake, in part because of public mistrust in government intentions and a lack of clear channels for engagement or recourse [34].

Together, these legal, institutional, and civic constraints form a comprehensive implementation gap, not due to a lack of technical vision, but due to asymmetrical governance structures that prioritize executive efficiency over horizontal accountability.

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5. Strengthening digital integrity: Pathways forward

Addressing the structural and legal gaps identified above requires more than incremental reforms. It calls for a shift toward systemic integrity-by-design, in which transparency and accountability are embedded into the architecture of digital governance from the outset. This section outlines four interlinked pathways to achieve that goal, rooted in comparative insights and grounded in the institutional realities of the region.

Reform in the Gulf must be sequenced strategically. In opaque and centralized systems, reform cannot begin with civic participation or judicial independence alone. It must begin by strengthening the connective tissue between technology, law, and oversight. Each pathway below is, therefore, designed not only as a standalone recommendation but as a building block within a broader ecosystem of governance renewal.

5.1 Embed transparency and accountability in digital systems

Digital platforms should be designed to integrate transparency features from inception – not added later as decorative elements. These include:

- Real-time red-flag alerts for procurement anomalies.
- Public audit trails with contract amendments and delivery benchmarks.

- Mandatory publication of procurement scoring criteria, and
- Open APIs allowing external actors to analyze data.

Examples from outside the region offer models for inspiration. Ukraine’s *ProZorro* platform requires all tender documents, evaluation criteria, and performance data to be published in real time, enabling civil society to monitor outcomes and detect irregularities [32, 33]. Similarly, *ChileCompra* incorporates social audits and dispute resolution mechanisms, enabling local communities to challenge procurement decisions.

In the GCC, platforms such as *Etimad* and *UAE Pass* could be upgraded to include layered transparency modules. For example, Etimad could:

- Require publication of final contract values and contractor performance,
- Disclose reasons for bid rejection,
- Integrate civil society “red-flag” dashboards, similar to those developed by TI Ukraine.

Such enhancements, however, must be backed by legal mandates that define the scope, governance standards, and oversight responsibilities of digital systems [9, 24]. Without this legal foundation, transparency features can be disabled, delayed, or politicized.

In short, digital architecture must be aligned with accountability architecture. Technologies must be intentionally designed not just for performance, but for public visibility, institutional linkage, and contestability.

AI and blockchain deployments in the public sector must also be covered by legislation that ensures algorithmic accountability, auditability, and citizen redress. As Albous et al. (2025) argue, national AI strategies must move beyond ethical statements to adopt concrete legal instruments that bind public agencies to standards of transparency, fairness, and due process [36].

Similarly, smart contracting tools based on blockchain must be supported by clear regulations on ledger governance, data retention, and access rights. Alshhadat (2024) recommends adopting regional interoperability protocols to facilitate data exchange across ministries while respecting data sovereignty and privacy concerns [34].

5.2 Strengthen legal frameworks and oversight institutions

Legal reforms are essential to enable meaningful digital accountability. FOI laws, where absent, should be introduced, and where they exist, they should be strengthened to include:

- Mandatory disclosure deadlines,
- Narrow and clearly defined exemptions (e.g., national security),
- Independent appeals mechanisms, and

- Protection of whistleblowers who disclose abuse.

In parallel, anti-corruption laws should be updated to incorporate digital integrity provisions, including:

- Mandatory digital disclosure of procurement outcomes,
- Algorithmic accountability clauses, and
- Institutional rights to audit blockchain or AI systems [23, 29].

At the institutional level, SAIs and anti-corruption agencies must be granted:

- Operational independence from the executive,
- Access to full digital data streams, and
- Powers to suspend, investigate, and refer suspicious digital transactions.

These reforms require more than administrative adjustments; they may entail:

- Constitutional amendments or legal reforms to clarify oversight mandates,
- Elite negotiation to protect institutional autonomy, and
- Pressure from international donors or trade partners, who increasingly integrate governance standards into agreements.

Examples from Morocco and Tunisia show that, even within constrained regimes, independent procurement courts and audit chambers can serve as catalysts for broader reform if given autonomy and international backing [27].

Even modest legal gains – such as requiring the publication of contract scoring matrices – can create precedents and alter institutional incentives.

5.3 Promote inclusive civic engagement and data literacy

Digital integrity is not just technical or legal – it is social and participatory. Citizens must have the knowledge, tools, and legal rights to access, understand, and act upon public data.

GCC governments should, therefore, invest in:

- Digital literacy campaigns focused on budget transparency and procurement oversight,
- Open data training programs in universities and schools of public administration, and
- Safe public engagement spaces, such as civic-tech labs and sanctioned hackathons.

One promising (if short-lived) example was Bahrain’s “OpenGov” initiative, which piloted participatory hackathons involving students, developers, and government officials [3, 14]. Such formats should be institutionalized, not abandoned.

In restrictive environments, indirect civic-tech ecosystems – such as university research labs, policy think tanks, or innovation hubs – can act as proxies for civil society [20]. International organizations, including UNDP, OECD, and the World Bank, can help establish such ecosystems through targeted grants and policy support.

Digital platforms should also include feedback interfaces: tools for flagging irregularities, submitting citizen reports, or requesting data corrections. Even asynchronous input, such as anonymous comment boxes or suggestion forums, can signal institutional responsiveness.

Ultimately, participation shifts expectations. Once citizens see their input lead to change – however incremental – they begin to perceive the state as responsive, creating a virtuous cycle of engagement and legitimacy.

5.4 Foster regional collaboration and benchmarking

Given their shared political economy and institutional similarities, GCC countries would benefit from regional cooperation on digital integrity standards. A Gulf Digital Integrity Index (GDII) – co-developed by independent institutions, audit bodies, and civic actors – could track progress in:

- e-Procurement transparency,
- Blockchain auditability,
- AI governance frameworks,
- Open data responsiveness, and
- Civic feedback channels.

Such a benchmarking initiative would serve multiple purposes:

- Create reputational incentives for reform through peer comparison,
- Expose blind spots or stagnation, and
- Enable cross-country learning, especially in tool design and legal integration.

Institutions such as the GCC Secretariat, INTOSAI, or the Arab Anti-Corruption and Integrity Network could host annual reviews, awards, and policy exchanges based on GDII performance [27].

International cooperation with entities such as OECD, UNODC, or the World Economic Forum could further anchor reform efforts by offering:

- Technical assistance on digital procurement,

- Ethical frameworks for AI use, and
- Capacity-building for data governance bodies [29, 32].

In the absence of strong domestic accountability, external benchmarking can act as a surrogate – not replacing national reform, but amplifying it through reputational leverage.

Ultimately, the pathways outlined here are not mutually exclusive, nor are they sufficient alone. The transformation of digital governance into a true anti-corruption instrument requires:

- Legal frameworks that enable transparency,
- Institutions that enforce it, and
- Citizens who activate it.

In contexts of limited political openness, progress may be incremental, fragmented, and contested. But if well-sequenced and grounded in local realities, it can be cumulative. The following conclusion reflects on the broader implications of such transformations.

6. Conclusion

Digital transformation is often presented as a silver bullet for governance challenges in the Gulf. While there is no doubt that e-government platforms, blockchain applications, and AI-driven oversight tools offer promising avenues to enhance public sector integrity, their true potential hinges on political will, legal grounding, institutional autonomy, and civic participation [1, 5, 9].

This chapter has shown that, across the GCC, digital innovations often function in technocratic silos – focused more on administrative efficiency than on systemic accountability. In many instances, digitalization has optimized state performance without disrupting the underlying asymmetries of power or information. As a result, public integrity gains remain limited, and the symbolic deployment of technology may mask enduring risks of elite capture, non-disclosure, and discretionary decision-making [12, 25].

The resulting implementation gap is not a failure of technology per se, but a reflection of governance priorities and institutional design. Digital tools are often deployed in ways that enhance executive coordination while avoiding structural reforms that would redistribute power or enable public contestation. The very systems meant to increase visibility are often shielded from scrutiny, and the platforms meant to facilitate participation remain inaccessible or irrelevant to ordinary citizens [6, 9, 38].

Yet, this gap is not immutable. International experience – as well as emerging dynamics within the Gulf – suggests that, under certain conditions, reform trajectories can be initiated. These include:

- The emergence of technocratic coalitions within ministries that recognize the reputational value of transparency,
- Pressure from trade partners, investors, or multilateral bodies linking digital performance to international rankings,
- Pilot reforms or digital sandboxes that create localized success stories, capable of inspiring gradual systemic uptake, and
- And, albeit more rare, civic-tech innovation hubs that navigate institutional boundaries through informal influence or soft diplomacy.

As argued throughout this chapter, the transformation of digital governance into a lever for public integrity requires a triangular convergence of law, technology, and civic oversight. Where only one of these pillars exists, progress will be partial and fragile. But where all three interact – within a sequenced and context-sensitive strategy – digital tools can become the foundation for a new social contract between states and citizens.

In the GCC context, this would mean moving from:

- Transparency by permission to transparency by default,
- Auditability in principle to auditability by law, and
- And, participation by invitation to participation by right.

Ultimately, what is at stake is not only the efficiency of digital platforms but the ethics of governance in the twenty-first century. Digital integrity is not simply about technological modernization; it is about redefining the moral architecture of public service, ensuring that digitization serves not just the state, but the public good.

If Gulf states can anchor their digital governance agendas in principles of legality, oversight, and inclusive design, they may not only improve procurement processes or service delivery but also strengthen institutional resilience, enhance legitimacy, and position themselves as pioneers of accountable digital government in the Global South.

Annex A – comparative table of digital integrity tools in the GCC

- Source: Author's own compilation.

This annex provides a comparative overview of digital anti-corruption tools implemented across six GCC countries, as referenced in Section 3 of this chapter.

Country	e-Procurement platform	Blockchain use	AI for risk detection	Transparency features	Public accessibility
Saudi Arabia	Etimad	Pilots in land and healthcare records	Tax data analytics	Centralized contracts portal	Partial access (registered users)
United Arab Emirates	Smart procurement under UAE Pass	Dubai Land block-chain registry	AI in customs and border control	Real-time public services portal	Public dashboard access
Qatar	e-Tendering via TASMU	Blockchain pilots for logistics	AI for fraud analytics	Integrated finance platforms	Limited external reporting
Kuwait	CAPT e-Procurement	Feasibility phase	No national deployment	Tender archives (partial)	Login required
Oman	e-Tendering System	Public registry integration under review	Audit tools via SAI	Basic contract publication	Accessible with registration
Bahrain	Government tender board system	eGov blockchain for docs	No disclosed AI use	Dashboard and summary reports	Publicly available

Annex B – legal and institutional enablers of digital integrity in the GCC

- Source: Author’s own compilation.

This annex summarizes the presence or absence of key legal frameworks and institutional mechanisms supporting digital integrity across the six GCC countries.

Country	FOI law in force	Independent oversight	Civic feedback mechanism	Procurement transparency mandate	AI/Blockchain legal framework
Saudi Arabia	✗ Absent	✗ Limited (SAI under executive)	✗ None	✗ Not legally required	✗ Fragmented
UAE	✗ Absent	✗ Executive supervision	✗ Minimal	✗ Discretionary	✗ Lacks AI regulation
Qatar	✓ Weak scope	✗ Partial independence	✗ Absent	✗ Non-binding strategy	✗ No oversight structure
Kuwait	✓ Partial	✓ Nazaha (limited power)	✗ No digital interface	✗ No legal mandate	✗ No audit mechanism
Bahrain	✓ Enacted	✗ Under council of ministers	✗ Limited	✓ Public portal	✗ Pilots only
Oman	✗ Absent	✗ Weak institutional support	✗ Absent	✓ Basic award data	✗ Early-stage exploration

Conflict of Interest

The authors declare no conflict of interest.

Acknowledgment

The author acknowledges the use of ChatGPT (OpenAI) for language polishing of the manuscript. The author retains full responsibility for the content and interpretation presented in this chapter.

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